

Activity Report

"Orientation on Hollow Concrete Block Manual"

HRRP Kathmandu Valley

19 March 2019

Activity Date and Venue:	19/03/2019, Nagarkot Hillside Village Resort, Bhaktapur district, State no. 3
Guests	Mahalaxmi Joshi (S.D.E , DLPIU chief, Bhaktapur), Narakanta Sapkota (M&E Consultant), Prabesh Adhikari (D.S.E, Bhaktapur), Ramesh Paudyal (MIS Officer, DLPIU Bhaktapur), Arun Adhikari (MIS Officer, Kavre)
Facilitator	Deepak Saud (HRRP), Sweata Sijapati (HRRP)
Report Prepared By:	Sweata Sijapati

Discussion:
Welcome Remarks Er. Prabesh Adhikari, D.S.E Bhaktapur welcomed all the participants who arrived in the Orientation Program on Hollow Concrete Block Manual. The orientation program was targeted for the technical staffs deployed by DLPIU. The objective of the orientation was to disseminate knowledge on the HCB manual and how it will help the technical staffs during the inspection of the house being constructed using such blocks. Remarks from DLPIU-Bhaktapur S.D.E Mahalaxmi Joshi, Chief, DLPIU- Building highlighted the importance of the published documents for the reconstruction process in all the district. She shared that lots of houses in Bhaktapur are being constructed using blocks as they are financially cheaper. Due to the such construction the non-compliance issues of such typologies are increasing. Finally, she mentioned that the published documents give flexibility to the reconstruction and many beneficiaries will be able to receive the grant and requested all the participants to be learn actively from this orientation program Orientation on Hollow Concrete Block Manual (Er. Deepak Saud, HRRP) The brief introduction to the Hollow Concrete Block structure, the background, objectives and content of the session along with its technical aspects, its applicability and construction practices were presented. Technical specifications for such shelter was discussed in brief. "Hollow Concrete Blocks Manual for Load Bearing Structures" for Reconstruction of Earthquake Resistant Houses, which has been published by the National Reconstruction Authority (NRA). This manual will support houses that have been constructing using concrete blocks in various district. The manual covers mainly inspection forms with comprehensive inspection methodology and cost-effective correction measures for non-compliant cases were suggested to guide Inspectors/Engineers while providing Social Technical Assistance (STA). The major non-compliance issues found in the HCB construction were quality of block units during production and the provision of seismic banding during house construction, on which performance of buildings is greatly rely.

Discussion:

The HCB manual will be used by all the engineers who are working for the construction and have been developed by the GoN to carryout inspections.

This manual has been divided into five parts and five annexes:

- PART-1: Background
- PART-2: Building Typology and Inspection
- PART-3: Technical Specification
- PART-4: Correction Measures
- PART-5: Ready to Use Design
- Annex A: HCB Unit Quality Test at Site
- Annex B: Structural Analysis and Design
- Annex C: Estimate of Correction Measures
- Annex D: Case Study on Inspection
- Annex E: Inspection Forms (HCB Masonry)

Nepal Standard: 119/2042 ensures the quality control requirements for HCBs production in Nepal, if followed properly. The Nominal block sizes of 400x200x200 mm, 400x150x200 mm, and 400x100x200 mm as per NS 119/042. Covers the reinforced masonry structural wall system building which are newly constructed or under construction using concrete blocks (Hollow or Solid) under the GoN housing reconstruction programme and is applicable for residential building and fall under category 'C' of NBC 2002:1994.

Inspection process of the HCB structure shall be based on specification provided in this manual hence, structural calculation is not required.

- Is applicable up to two storey, ground floor with masonry (SMC, BMC ONLY) structure and first floor with Hollow Concrete Block structure.
- Up to two storey, both ground and first floor with Hollow Concrete Block

However, if the building is more than two storey, structural calculation is mandatory.

The ready to use design included in the manual are applicable for following structure:

- Number of storey: 2
- Storey height: 2.50 m (maximum)
- Total height: 5.00m
- Unsupported wall length: 4.05 m (maximum)
- Plinth area: 100 sq.m.

All the details of the construction process, the quality test of the block in field along with the details of the wall test and individual block test conducted in Institute of Engineering (IOE) is included in the manual. The manual has been prepared in such a way that it can be easily understood and can be applicable for all the structure made of blocks.

Questions Collected	Who asked the Question?	Response Provided
Which value of the compressive strength was taken in the calculation of the 2 storey HCB block structure?	DLPIU	More than 80% of the blocks collected from the field from 14 districts were found of poor quality. So, for the calculation minimum allowable value of the compressive strength of the block was considered.
How do we conduct the quality test of the already finished house?	Engineer, POs	It might be difficult to do the quality test for already completed structure. If there are remaining block from the construction, then the quality test can be done as indicated in the manual. If there are no leftover block, then with the approval from the beneficiary the block from the top level of the house below the roof can be removed to conduct the test.
Does this manual only cover the house build using the Hollow blocks?	Engineer, POs	Covers the reinforced masonry structural wall system building which are newly constructed or under construction using concrete blocks (Hollow or Solid) under the GoN housing reconstruction programme.

Photographs



Participants in the orientation program on Hollow Concrete Block manual conducted in Bhaktapur district.



DMT Technical Coordination Officer, HRRP is conducting technical session on Hollow Concrete Block manual

Attendance						
Housing Recovery and Reconstruction Platform (HRRP)						
HCB Orientation, Bhaktapur						
Date: 19-Mar-19						
Venue: Nagarkot Hillside Village Resort, Nagarkot, Bhaktapur						
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